

Risk-Averse Evaluation of Conformal Scores through Conditional Value-at-Risk

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Abstract

Conformal prediction provides distribution-free guarantees by constructing prediction sets that contain the true outcome with a predefined probability, under the mild assumption of exchangeability. While this marginal coverage guarantee is powerful, it remains an aggregate measure, by controlling the frequency of errors but being agnostic to their severity.

In many high-stakes applications, however, not all errors are equally critical. Two conformal predictors achieving the same nominal coverage may exhibit different tail behaviors: one may incur into large deviations from the empirical quantile, while another may keep closer to it. However, standard evaluation metrics, such as empirical coverage and average set size, fail to capture this aspect.

To address this limitation, we propose to evaluate conformal predictors through the lens of *Conditional Value-at-Risk*, a well-established measure of tail risk. This perspective enables a risk-averse selection of score functions, favoring those that control not only the frequency but also the magnitude of errors. In particular, we select the score function that minimizes this tail risk, thereby ensuring a tighter alignment between the worst errors observed on test data and the quantile estimated during calibration.

Experiments across multiple datasets and classifiers show that our criterion captures risk profiles that are not revealed by coverage and average set size alone, often favoring more conservative score functions and exposing differences between scores that appear equivalent under canonical metrics.

Keywords: Conformal prediction; conditional value at risk; decision making; trustworthy artificial intelligence; safety-critical applications.

1. Introduction

Conformal prediction (CP) has become a central framework for distribution-free uncertainty quantification in machine learning (ML), as it provides finite-sample marginal coverage guarantees under the mild assumption of exchangeability while remaining model-agnostic and broadly applicable across data modalities (Shafer and Vovk, 2008). In classification, CP returns prediction sets that contain the true label with a user-defined probability. Such a guarantee is intrinsically marginal because it controls the frequency of miscoverage events, but

it does not provide any information on their severity, namely, the order of magnitude of the errors. As a consequence, two conformal predictors may achieve similar empirical coverage while exhibiting markedly different behavior in the tail, with one predictor producing several mild coverage violations and another incurring rare but severe violations. A key factor influencing this behavior is the choice of the nonconformity score function. This score determines the structure of the resulting conformal sets according to its underlying design mechanism. Depending on the degree of conservatism embedded in their definitions, different scores may induce smaller or larger prediction sets, and therefore lead to substantially different sets even when applied over the same collection of data.

The comparison of score functions is usually performed through the average set size, once that the marginal coverage is satisfied. This may favor compact, minimum-size, prediction sets, but does not explicitly account for the severity of the errors. Also, in case of ties, that is when different score functions achieve comparable set size, the choice of the best one is typically found via heuristic approaches, such as ad-hoc experiments and domain expertise; the latter further introducing a degree of empiricism. This limitation is especially relevant for safety-critical scenarios with conformal errors having different amplitudes (Seferis et al., 2023). A CP method that fails rarely, but severely, may be less desirable than one whose failures are more controlled, even if both achieve comparable coverage and average set size. Standard CP metrics, however, do not explicitly capture this distinction. Recent work on CP for non-stationary time series highlights this broader gap, noting the lack of formal tools to statistically evaluate conformal sets (Retzlaff et al., 2025).

A central question then arises: *how can we compare score functions in terms of the severity of the conformal errors they induce?* We investigate this problem by exploiting Conditional Value-at-Risk (CVaR) (Rockafellar and Uryasev, 2000), a classical coherent measure of tail risk (Artzner et al., 1999). We demonstrate how it can be used to establish a new risk-averse criterion for the evaluation and selection of CP score functions. Specifically, by computing CVaR to a conformal loss that measures the excess of the true-class score over the empirical quantile, we obtain a metric that captures the severity of miscoverage events. This way, we add error severity as a third dimension for CP scores evaluation beyond standard coverage and size. Moreover, we show how this strategy drives appropriate selection even when state-of-the-art approaches do not.

The remaining of the paper is organized as follows: Section 2 presents relevant literature related to our work; Section 3 describes the fundamental concepts of CP for classification models, and introduces the main definitions of CVaR; Section 4 reports the core of our work, describing the role of CVaR in CP evaluation and selection; Section 5 describes the experiments and results obtained to evaluate our approach; finally, Section 6 concludes the paper and reports some future works on the topic.

2. Related work

In this section, we review related work on CP evaluation beyond coverage guarantees, with a focus on risk-aware approaches and their role in high-stakes decision making.

2.1. Conformal prediction evaluation beyond coverage guarantees

Following its foundational formulation (Shafer and Vovk, 2008), CP provides prediction sets satisfying the *marginal coverage guarantee* (or *validity*), ensuring that the true outcome is contained in the set with a predefined probability.

Beyond validity, CP evaluation is typically characterized by two main axes: *efficiency* and *adaptivity*. Efficiency measures how compact prediction sets are on average while maintaining validity, and is commonly quantified via average set size or related statistics such as the frequency of singleton and multi-element sets (Vovk et al., 2016). More recent work has also linked efficiency to properties of the underlying predictive model, showing connections between generalization and prediction set size (Zecchin et al., 2024).

Adaptivity captures the ability of prediction sets to adjust their size according to input difficulty or uncertainty. It is typically assessed through stratified or conditional evaluations, such as size-conditioned coverage violations or subgroup-based analyses (Angelopoulos et al., 2020; Huang et al., 2024), as well as input-transformation-based metrics such as transformation-based binning coverage violation and transformation-based binning set size (Jang and Lee, 2026). Complementary to these notions, Linusson et al. (2020) study *p-value stability* under different calibration splits in inductive CP, highlighting variability as an additional dimension of evaluation beyond validity and efficiency. Furthermore, another recent work on False Coverage Proportion control (Blain et al., 2025) strengthened marginal conformal validity by controlling the realized proportion of miscovered test points.

Despite these advances, most standard metrics capture only average-case behavior and do not explicitly reflect the severity or structure of prediction errors. This limitation has motivated a growing body of work on risk-aware extensions of CP. Early approaches include CP with limited false positives (Fisch et al., 2022; Carlevaro et al., 2024), which explicitly constrains the inclusion of incorrect labels in prediction sets, improving selectivity while preserving validity. A more general framework is provided by Risk-Controlling Prediction Sets (Bates et al., 2021), which guarantees satisfaction of user-specified risk constraints.

Building on this idea, Conformal Risk Control (Angelopoulos et al., 2022) replaces coverage constraints with constraints on the expected value of a user-defined loss, enabling flexible control over different types of errors beyond miscoverage. Several extensions refine this framework further. Conformal Loss-Controlling Prediction (Wang et al., 2024) enforces constraints on individual loss realizations rather than only expectations, while Selective Conformal Risk Control introduces a two-stage procedure combining sample selection and risk control (Xu et al., 2025). The Learn-Then-Test framework (Angelopoulos et al., 2025) provides an alternative perspective by treating risk control as a model selection problem over candidate predictors tested against calibration-based statistical guarantees.

Recent work has also begun to explore more general risk measures beyond expectation. In particular, Yeh et al. (2023) extends Conformal Risk Control (Angelopoulos et al., 2022) to Optimized Certainty Equivalent risks, a broad class that includes both expected loss and tail-sensitive measures such as CVaR, thereby enabling explicit control of worst-case outcomes in conformal prediction.

2.2. Conformal prediction for risk-sensitive decision making

Beyond uncertainty quantification, CP has increasingly been studied from a decision-theoretic perspective, where prediction sets are used to support downstream actions under uncertainty. A first step in this direction is provided by [Vovk and Bendtsen \(2018\)](#), who introduce conformal predictive decision making, where prediction sets are directly embedded into decision procedures, showing how conformal guarantees can guide optimal actions under uncertainty.

A fundamental contribution is given by [Kiyani et al. \(2025\)](#), which establishes a decision-theoretic foundation for CP under risk-averse objectives. The authors show that prediction sets are optimal uncertainty representations for reinforcement-learning agents optimizing Value-at-Risk (VaR), and characterize optimal decision policies under such risk constraints. This result formally connects CP with risk-sensitive decision theory, framing prediction sets as tools for decision optimization rather than purely statistical guarantees.

This perspective has been further extended in several directions. [Wu et al. \(2026\)](#) interpret CP as a decision risk problem in a Bayesian framework, strengthening the connection between conformal methods and principled decision-theoretic uncertainty quantification. In a similar spirit, [Wang and Dobriban \(2026\)](#) propose Risk-Optimal CP, which formulates prediction set construction as an optimization problem over downstream decision loss, aiming to minimize expected risk rather than solely enforce coverage.

Recent work further emphasizes the role of utility and decision objectives in conformal methods. In particular, [Cortes-Gomez et al. \(2024\)](#) introduce a utility-directed framework in which prediction sets are explicitly optimized for downstream utility, while [Lekeufack et al. \(2024\)](#) develop a conformal decision theory framework for safe autonomous decision-making, integrating conformal sets into decision pipelines with safety guarantees.

Finally, a closely related line of work studies selective classification and reject options, where models are allowed to abstain when uncertainty is high ([Szabadvary et al., 2025](#); [Fang and Nalisnick, 2024](#); [Kumar et al., 2025](#); [Boursinos and Koutsoukos, 2022](#)). These approaches highlight how CP can support abstention-based decision strategies, reducing the cost of incorrect predictions in high-stakes settings.

3. Background concepts

Before detailing our theoretical contributions, we provide the necessary background on inductive Conformal Prediction (CP) and Conditional Value-at-Risk (CVaR). This section also introduces the notation and core terminology that form the foundation of our analysis.

3.1. Inductive conformal prediction

In this paper, we refer to multiclass classification models learning a mapping function $f : \mathcal{X} \rightarrow \mathcal{Y}$, from a D -dimensional feature space $\mathcal{X} \subseteq \mathbb{R}^D$ to a label space $\mathcal{Y} = \{1, \dots, K\}$ with $K \geq 2$ classes. We denote with $\mathbb{f}(\mathbf{x})_y$ the model predicted probability for sample $\mathbf{x}$ and label $y \in \mathcal{Y}$.

We consider the inductive CP setting ([Shafer and Vovk, 2008](#)). A dataset $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ of labeled, exchangeable points (i.e., whose joint probability distribution is invariant under permutations) is partitioned into a proper training set

$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_{n_t}, y_{n_t})\}$, used to fit the classifier f , and an independent calibration set $\mathcal{D}_{\text{cal}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_{n_c}, y_{n_c})\}$, used to estimate the predictor’s uncertainty, such that $n = n_t + n_c$ and $\mathcal{D}_{\text{train}} \cap \mathcal{D}_{\text{cal}} = \emptyset$. Then a test set $\mathcal{D}_{\text{test}}$ is drawn independently from $\mathcal{D}$. Model uncertainty is encoded by a negative-oriented (i.e., higher, more uncertain) non-conformity *score function* $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$, which, given a user-defined error level $\alpha \in (0, 1)$, is used to compute the $\lceil (n_c + 1)(1 - \alpha) \rceil / n_c$ -empirical quantile of the score values in the calibration set

$$q_{1-\alpha}(\mathbf{s}) \doteq \min \left[q : q \in \mathbb{R}, \frac{1}{n_c} \sum_{i=1}^{n_c} \mathbb{1}\{s_i \leq q\} \geq 1 - \alpha \right], \quad (1)$$

where $\mathbf{s} = \{s_i\}_{i=1, \dots, n_c}$, $s_i = s(\mathbf{x}_i, y_i)$.

Then, for a test sample $(\mathbf{x}_{n+1}, y_{n+1})$, the CP procedure constructs a conformal set

$$\mathcal{C}_\alpha(\mathbf{x}_{n+1}) \doteq \{y \in \mathcal{Y} : s(\mathbf{x}_{n+1}, y) \leq q_{1-\alpha}(\mathbf{s})\} \in 2^{\mathcal{Y}}, \quad (2)$$

satisfying the marginal coverage guarantee at level of confidence $1 - \alpha$, i.e.,

$$\mathbb{P}\{y_{n+1} \in \mathcal{C}_\alpha(\mathbf{x}_{n+1})\} \geq 1 - \alpha. \quad (3)$$

Several score functions have been defined for classification tasks. In our work, we will consider the following ones:

- **Least Ambiguous set-valued Classifier (LAC)**, $s(\mathbf{x}, y) = 1 - \mathbb{f}(\mathbf{x})_y$, where $\mathbb{f}(\mathbf{x})_y$ is the class y probability for sample $\mathbf{x}$ (Sadinle et al., 2019). It prioritizes selecting the most confident predictions by assigning lower scores to labels with higher (probabilistically estimated) model confidence: the lower the score the more confident the model in its prediction;
- **Margin**, $s(\mathbf{x}, y) = \max_{y' \neq y} \mathbb{f}(\mathbf{x})_{y'} - \mathbb{f}(\mathbf{x})_y$. This score measures how much the most competitive alternative label y' exceeds the score of the true label y . Negative values indicate larger confidence in y , while larger (positive) values indicate ambiguity or potential misclassification (Johansson et al., 2013);
- **Adaptive Prediction Set (APS)**, $s(\mathbf{x}, y) = \sum_{i=1}^{k(y)} \mathbb{f}(\mathbf{x})_{\pi_i}$, where π is a permutation of the label indices such that $\mathbb{f}(\mathbf{x})_{\pi_1} \geq \mathbb{f}(\mathbf{x})_{\pi_2} \geq \dots$, and $k(y)$ is the rank of label y in this ordering (i.e., $\pi_{k(y)} = y$), (Romano et al., 2020; Angelopoulos et al., 2020). This ensures prediction sets that are, by construction, non-empty.
- **Regularized Adaptive Prediction Set (RAPs)** (Angelopoulos et al., 2021), has been introduced as a more efficient variant of APS method. The non conformity scores are computed by summing the regularized ranked scores of each label, from the higher to the lower until reaching the true label of the observation. Let π be a permutation of the label indices such that $\mathbb{f}(\mathbf{x})_{\pi_1} \geq \mathbb{f}(\mathbf{x})_{\pi_2} \geq \dots \geq \mathbb{f}(\mathbf{x})_{\pi_K}$, and let $k(y)$ denote the rank of label y , i.e., $\pi_{k(y)} = y$. The RAPs score is then defined as $s(\mathbf{x}, y) = \sum_{j=1}^{k(y)} \mathbb{f}(\mathbf{x})_{\pi_j} + \lambda[k(y) - k_{\text{reg}}]_+$, where $[z]_+ = \max\{z, 0\}$, $\lambda \geq 0$ is a regularization parameter, and k_{reg} controls the rank after which the penalty is applied. The regularization term penalizes labels appearing deep in the ranked probability list, thereby discouraging excessively large prediction sets.

3.2. Conditional Value-at-Risk

Conditional Value-at-Risk (CVaR), also known as expected shortfall, is a popular tail risk measure (Rockafellar and Uryasev, 2000), introduced for the economics field, and later expanded to other applications, including statistical learning (Soma and Yoshida, 2020). Given a real-valued loss random variable L , and a level $\beta \in (0, 1)$, CVaR captures the expected magnitude of L in the worst β -fraction of outcomes, i.e., the upper tail of the distribution with probability mass β . Specifically, it is defined (when the conditional expectation is well-defined) as:

$$\text{CVaR}_\beta(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\beta(L)], \quad (4)$$

where $\text{VaR}_\beta(L)$ is the Value-at-Risk of the variable at level β , which is its $(1 - \beta)$ quantile:

$$\text{VaR}_\beta(L) = \inf \{t \in \mathbb{R} : \mathbb{P}(L \leq t) \geq 1 - \beta\}. \quad (5)$$

Unlike VaR, which only captures a quantile of the loss distribution, CVaR naturally captures the scenarios in which a large loss is encountered. CVaR belongs to coherent risk measures (Delbaen, 2002): it satisfies properties such as monotonicity, subadditivity, positive homogeneity, and translation invariance, making it well-suited for risk-sensitive optimization. An equivalent and widely used characterization of CVaR is given by the variational form:

$$\text{CVaR}_\beta(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{\beta} \mathbb{E}[[L - \tau]_+] \right\}, \quad (6)$$

where $[u]_+ = \max\{u, 0\}$ and τ is a threshold level penalizing the losses above it. At the optimal, it holds that $\tau^* = \text{VaR}_\beta$. This formulation is particularly useful for providing an empirical estimation of CVaR given on a finite set of loss realizations $\{L_i\}_{i=1}^n$, that is:

$$\widehat{\text{CVaR}}_\beta(L) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{n\beta} \sum_{i=1}^n [L_i - \tau]_+ \right\}. \quad (7)$$

3.2.1. CVAR IN MACHINE LEARNING

In recent years, CVaR has gained increasing interests in the context of statistical and machine learning, where it is primarily used within novel optimization frameworks (Soma and Yoshida, 2020; Filippi et al., 2020) that move beyond average performance and explicitly control worst-case outcomes, especially when rare but severe errors matter. Early works connected CVaR to robust learning by reformulating classical models such as single-class support vector machines to limit extreme losses, improving stability under uncertainty (Takeda and Kanamori, 2009; Wang et al., 2015). More recent approaches integrate CVaR directly into learning algorithms: for instance, boosted CVaR classification replaces standard empirical risk with a tail-focused objective, leading to classifiers that are less sensitive to noise and hard examples (Zhai et al., 2021). On the theoretical side, PAC-Bayesian bounds for CVaR clarify how focusing on worst-case losses affects generalization, providing guarantees analogous to those available for expectation-based learning (Mhammedi et al., 2020), while adaptive sampling methods have been proposed to make CVaR optimization computationally feasible when tail events are rare (Curi et al., 2020). CVaR also plays a central role in dynamical

systems: it has been used to analyze and bound the risk of unsafe events in dynamical systems (Chapman et al., 2021) and to define constraints for safe reinforcement learning that directly limit the severity of high-cost trajectories during policy optimization (Zhang et al., 2025). Beyond safety, CVaR has been adopted as an evaluation and design tool for broader system objectives, including fairness, where it emphasizes the worst-performing groups rather than overall averages (Paes et al., 2024), explainable motion planning, where it quantifies extreme trajectory risks (Luo et al., 2025), and financial decision-making, where machine learning models are trained to respect CVaR-based risk parity criteria (Sheng et al., 2025). Together, these contributions show how CVaR provides a simple and flexible way to encode risk sensitivity across a wide range of machine learning problems. However, CVaR connections with CP theory remain little explored, although CP is one of the most established uncertainty techniques available today.

4. Evaluating conformal sets via Conditional Value-at-Risk

In this section, we present our proposed approach to blend CVaR and CP assessment under a risk-averse perspective.

Given a family $\mathcal{S}$ of score functions s , with the characteristics described in Sec. 3.1, we introduce a novel criterion for their evaluation that looks at how the score behaves in worst-case scenarios, i.e., in the up to α -fraction of samples, corresponding to potential violations of marginal coverage. We show how CVaR provides a straightforward way to achieve this goal. Since CVaR is sensitive to scale by definition (Eq. (4)), we unified scores so to vary in the $[0, 1]$ range, by applying min-max scaling, using the transformation parameters obtained from calibration data for both calibration and test scores normalization.

Definition 1 (Conformal Loss Function) *Given a score function s , we define its associated conformal loss function $L_s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$, as:*

$$L_s(\mathbf{x}, y) \doteq [s(\mathbf{x}, y) - q_{1-\alpha}(\mathbf{s})]_+, \quad (8)$$

where $s(\mathbf{x}, y)$ is the score value associated to sample $\mathbf{x}$ with ground truth label y , $q_{1-\alpha}(\mathbf{s})$ is the calibration quantile from (1), and $[\cdot]_+ = \max\{0, \cdot\}$.

This function encodes the severity of prediction set errors, as captured by the distance between score values and the calibration empirical quantile. In practice, L_s is a ReLU-like loss, yielding $L_s(\mathbf{x}, y) = 0$ when $y \in \mathcal{C}_\alpha(\mathbf{x})$, that is, when marginal coverage is satisfied, and $L_s > 0$ otherwise, thus quantifying the magnitude by which the score associated to the true label exceeds the conformal threshold.

CVaR-guided score selection. The best score function s_{RA}^* under a risk-averse perspective is the one obtained through CVaR by solving:

$$s_{\text{RA}}^* = \arg \min_{s \in \mathcal{S}} \text{CVaR}_\beta(L_s), \quad (9)$$

provided the satisfaction of the validity guarantee (3).

Given the availability of a test set $\mathcal{D}_{\text{test}}$, $\text{CVaR}_\beta(L_s)$ will be empirically computed following its empirical formulation (7), over the values of L_s across the test set.

4.1. Relationship between α and β

In principle, for computing $\text{CVaR}_\beta(L_s)$ ((4),(6)), one may consider any $\beta \in (0, 1)$ independently of CP error level α , thereby letting CVaR be calculated as the average of the worst β -fraction of L_s samples.

However, here we prove that, under our L_s definition, setting $\beta = \alpha$ formally establishes a connection between CP sets miscoverage severity and CVaR, allowing us to use CVaR as a miscoverage severity metric. Specifically, we demonstrate that when $\beta = \alpha$, CVaR provides (i) a pure measure of miscoverage severity under exact calibration, and (ii) a joint measure of miscoverage severity and calibration conservativeness otherwise.

First, provided that a CP set is valid (i.e., it satisfies the marginal coverage property), using $\beta = \alpha$ individuates a fixed value for VaR, i.e., $\text{VaR}_\alpha(L_s) = 0$, as formalized in Theorem 2.

Building on this, we show that under the exact coverage assumption (i.e., ideal calibration, expressed by the equality in (2)), choosing $\beta = \alpha$ implies that CVaR value arises exclusively from positive losses, i.e. from those samples contributing to the α rate of miscoverage events, as proved in Theorem 3.

Under the more relaxed, and realistic, assumption of a non-ideal – yet satisfied – marginal coverage (i.e., in the cases where exceedance in (2) holds), choosing $\beta = \alpha$ makes CVaR also reflect the conservativeness of the CP set. Let p_{mis} be the empirical miscoverage rate. Under non-ideal marginal coverage, p_{mis} is smaller than the nominal level α . Equivalently, the predictor exhibits a positive coverage gap, equal to $\alpha - p_{\text{mis}}$. In this setting, the worst α -fraction of the loss distribution contains all samples with positive losses (i.e., a p_{mis} fraction), corresponding to actual miscoverage events, together with an additional $(\alpha - p_{\text{mis}})$ -fraction of correctly covered samples whose loss value is zero. Hence, $\text{CVaR}_\alpha(L_s)$ no longer coincides only with the average severity of miscoverage events, but it also reflects the conservativeness of the CP set through a scaling factor p_{mis}/α . This is formally stated in Corollary 4.

In contrast, choosing $\beta > \alpha$ would not be semantically meaningful from a risk-averse score benchmarking perspective, as the proportion of zero-loss points included in CVaR computation would further increase, with CVaR still being affected by the severity of miscoverage events, but this contribution would be weakened by the larger fraction of correctly covered samples in the β -tail of L_s distribution.

On the other hand, choosing $\beta < \alpha$ can be meaningful when the objective is to focus on a more selective portion of the loss tail. Under exact coverage, where $p_{\text{mis}} = \alpha$, selecting $\beta < \alpha$ makes $\text{CVaR}_\beta(L_s)$ depend only on the largest positive losses, i.e., on the most severe miscoverage events. In this case, the corresponding VaR threshold moves inside the positive-loss region, so that $\text{CVaR}_\beta(L_s)$ no longer averages all miscoverage events, but only their worst β -fraction. Under non-exact marginal coverage, where $p_{\text{mis}} < \alpha$, the interpretation of $\beta < \alpha$ depends on its relation with p_{mis} . If $p_{\text{mis}} < \beta < \alpha$, the CVaR tail still contains all positive-loss samples, but fewer zero-loss samples than in the case $\beta = \alpha$. Therefore, $\text{CVaR}_\beta(L_s)$ remains a joint measure of miscoverage severity and conservativeness, with a reduced weakening due to zero-loss samples. Equivalently, an appropriate scaling factor (shown later in (17)) becomes p_{mis}/β , which is larger than p_{mis}/α and therefore closer to 1. If $\beta = p_{\text{mis}}$, CVaR coincides with the average severity of the realized miscoverage events. Finally, if $\beta < p_{\text{mis}}$, CVaR focuses only on the most severe subset of miscoverage events, thus providing a more extreme tail-risk assessment.

Theorem 2 (Value-at-Risk of L_s at $\beta = \alpha$) *Let $\mathcal{C}_\alpha(\mathbf{x})$ be a conformal prediction set satisfying marginal coverage guarantee. Let s be a score function and L_s its associated conformal loss function (8). Then, the following holds*

$$\beta = \alpha \implies \text{VaR}_\alpha(L_s) = 0. \quad (10)$$

Proof

Given the definition of L_s as per (8), considering a prediction set $\mathcal{C}_\alpha(\mathbf{x})$ satisfying the marginal coverage guarantees, it holds that

$$\mathbb{P}\{y \in \mathcal{C}_\alpha(\mathbf{x})\} = \mathbb{P}\{L_s = 0\} \geq 1 - \alpha. \quad (11)$$

Hence, using $\beta = \alpha$ in (5), the smallest value t such that $\mathbb{P}\{L_s \leq t\} \geq 1 - \alpha$ is $t = 0$. Therefore, it is

$$\text{VaR}_\beta(L_s) = 0. \quad (12)$$

■

Theorem 3 (CVaR at level α under exact coverage) *Let $C_\alpha(\mathbf{x})$ be a CP set satisfying exact marginal coverage. Let s be a score function and let L_s (8) be the associated conformal loss. Then, by choosing $\beta = \alpha$, it holds that*

$$\text{CVaR}_\alpha(L_s) = \frac{1}{\alpha} \mathbb{E}[L_s] = \mathbb{E}[L_s \mid L_s > 0]. \quad (13)$$

Hence, $\text{CVaR}_\alpha(L_s)$ captures the expected severity of miscoverage events.

Proof By definition of the conformal loss,

$$L_s(\mathbf{x}, y) = 0 \iff s(\mathbf{x}, y) \leq q_{1-\alpha}(s) \iff y \in C_\alpha(\mathbf{x}),$$

whereas

$$L_s(\mathbf{x}, y) > 0 \iff s(\mathbf{x}, y) > q_{1-\alpha}(s) \iff y \notin C_\alpha(\mathbf{x}).$$

Under the exact coverage assumption,

$$\mathbb{P}\{y \in C_\alpha(\mathbf{x})\} = 1 - \alpha,$$

and therefore

$$\mathbb{P}\{L_s = 0\} = 1 - \alpha, \quad \mathbb{P}\{L_s > 0\} = \alpha. \quad (14)$$

Now, using the variational formula for CVaR (6) with $\beta = \alpha$, we have

$$\text{CVaR}_\alpha(L_s) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \frac{1}{\alpha} \mathbb{E}[[L_s - \tau]_+] \right\}.$$

Since $\tau^* = \text{VaR}_\alpha(L_s)$ and $\text{VaR}_\alpha(L_s) = 0$ from Theorem 2, this formulation reduces to

$$\text{CVaR}_\alpha(L_s) = \frac{1}{\alpha} \mathbb{E}[L_s], \quad (15)$$

where the expectation can be decomposed as

$$\mathbb{E}[L_s] = \underbrace{\mathbb{P}\{L_s = 0\}}_{1-\alpha} \underbrace{\mathbb{E}[L_s \mid L_s = 0]}_0 + \underbrace{\mathbb{P}\{L_s > 0\}}_{\alpha} \mathbb{E}[L_s \mid L_s > 0],$$

which then simplifies to

$$\mathbb{E}[L_s] = \alpha \mathbb{E}[L_s \mid L_s > 0].$$

Substituting it into Eq.(15) it yields

$$\text{CVaR}_\alpha(L_s) = \mathbb{E}[L_s \mid L_s > 0].$$

■

Corollary 4 (CVaR at level α under non-exact coverage) *Let $C_\alpha(\mathbf{x})$ be a conformal prediction set satisfying the marginal coverage guarantee with strict inequality, i.e.,*

$$\mathbb{P}\{y \in C_\alpha(\mathbf{x})\} > 1 - \alpha. \quad (16)$$

Let s be a score function and L_s be the associated conformal loss (8). Denote by p_{mis} the miscoverage rate. If $p_{\text{mis}} < \alpha$, then

$$\text{CVaR}_\alpha(L_s) = \frac{p_{\text{mis}}}{\alpha} \mathbb{E}[L_s \mid L_s > 0]. \quad (17)$$

That is, $\text{CVaR}_\alpha(L_s)$ jointly reflects the severity of the observed miscoverage events and the conservativeness of the conformal predictor through a scaling factor p_{mis}/α .

Proof By definition of the conformal loss,

$$L_s(\mathbf{x}, y) = 0 \iff s(\mathbf{x}, y) \leq q_{1-\alpha}(\mathbf{s}) \iff y \in C_\alpha(\mathbf{x}),$$

whereas

$$L_s(\mathbf{x}, y) > 0 \iff s(\mathbf{x}, y) > q_{1-\alpha}(\mathbf{s}) \iff y \notin C_\alpha(\mathbf{x}).$$

Under the non-exact coverage hypothesis

$$p_{\text{mis}} = \mathbb{P}\{y \notin C_\alpha(\mathbf{x})\} = \mathbb{P}\{L_s > 0\} < \alpha,$$

and consequently

$$\mathbb{P}\{L_s = 0\} = 1 - p_{\text{mis}} > 1 - \alpha.$$

From Theorem 2

we obtain again the expression of (15), where the expectation can be decomposed as

$$\mathbb{E}[L_s] = \underbrace{\mathbb{P}\{L_s = 0\}}_{>1-\alpha} \underbrace{\mathbb{E}[L_s \mid L_s = 0]}_0 + \underbrace{\mathbb{P}\{L_s > 0\}}_{p_{\text{mis}}} \mathbb{E}[L_s \mid L_s > 0],$$

which this time simplifies to

$$\mathbb{E}[L_s] = p_{\text{mis}} \mathbb{E}[L_s \mid L_s > 0].$$

Substituting it into Eq. (15) yields

$$\text{CVaR}_\alpha(L_s) = \frac{p_{\text{mis}}}{\alpha} \mathbb{E}[L_s \mid L_s > 0].$$

■

5. Experiments

This section presents the experiments and discusses the obtained results of our CVaR-based approach to CP evaluation. After describing the adopted datasets and classifiers in Section 5.1, we illustrate the outcomes of our proposed method in 5.2. Finally, we summarize and discuss the main findings in 5.3.

We ran all experiments using Python3.12 on a machine equipped with an Apple M4 CPU, 24 GB RAM, running MacOS 26.3.1. The code is available for reproducibility at the following GitHub repository: <https://github.com/saranrt95/CVAR-CP>.

5.1. Datasets and baseline models

The proposed method has been tested on 10 widespread classification datasets available from UCI Machine Learning repository ¹, with the characteristics summarized in Table 1. Notably, these datasets differ both in terms of sample size, dimensionality, and number of classes.

Dataset	UCI ID	# samples	# features	# classes	Test Accuracy [%]		
					RF	SVM	MLP
breastW	17	569	30	2	97	98	97
digits	80	5620	64	10	97	99	98
drybean	602	13611	16	7	92	93	93
liver	225	583	10	2	68	74	70
spambase	94	4601	57	2	95	93	94
st_image	147	2310	18	7	96	93	95
st_landsat	146	6435	36	6	90	91	90
waveform	107	5000	21	3	84	86	85
wine	186	4898	11	7	66	58	56
yeast	110	1484	8	10	60	57	60

Table 1: Datasets from UCI repository used in this paper.

We first standardized the data using z-score transform, and then split them into a 50% portion for proper training set, a 30% for calibration set, and the remaining 20% for test set.

1. <https://archive.ics.uci.edu/>

Then, we trained three classifiers on each training dataset, namely, a random forest (RF), a support vector machine (SVM) with radial basis function kernel, and a multi-layer perceptron. We selected the hyperparameters through a 3-fold cross-validation by searching within the following configurations, as per the scikit-learn Python library implementation (Pedregosa et al., 2011):

- RF: `n_estimators` $\in \{100, 300, 500\}$; `criterion` $\in \{gini, entropy\}$;
- SVM: `C` $\in \{0.001, 0.01, 0.1, 1, 10, 100\}$;
- MLP: `learning_rate_init` $\in \{0.001, 0.0001\}$; `hidden_layer_sizes` $\in \{(128, 64), (128, 64, 32), (64, 32, 16), (32, 16), (64, 32)\}$; `max_iter` $\in \{200, 500, 1000\}$.

After setting the optimal parameters, the final classifiers have been retrained on the whole training set, achieving the accuracy on test data as shown in the last column of Table 1.

5.2. Evaluation results

Once the models have been trained, we carried out the inductive CP pipeline using LAC, Margin, APS and RAPS score functions (see 3.1). The empirical quantiles, at a given level α , from calibration samples were used to calculate the loss values as per Eq. (8), from which the CVaR-based analysis has been conducted.

In the following Sections, we first provide empirical results supporting Theorem 2, and report an illustrative example to show the link between score and loss functions distributions when $\beta = \alpha$. Then, we compare the four score functions through canonical metrics versus CVaR. Lastly, we conduct a pairwise score analysis on the outcomes obtained at the previous evaluation stage, showing that CVaR can drive the choice of a ‘winner’ score function even when canonical metrics are not enough to reveal significant differences between the compared scores.

5.2.1. EXPERIMENTAL ANALYSIS OF VaR_α

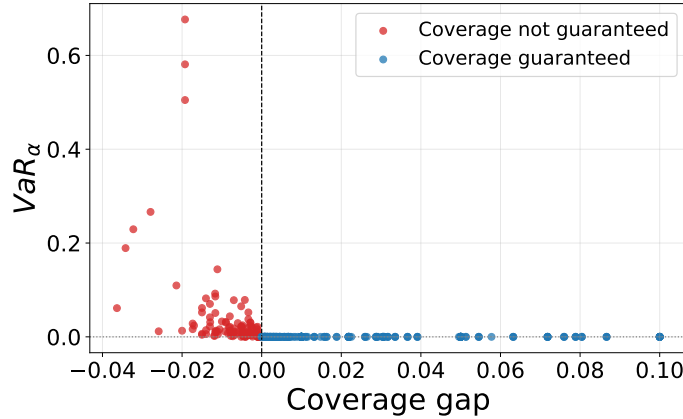


Figure 1: Illustration of the relationship between $\text{VaR}_\alpha(L_s)$ and marginal validity, where the latter is assessed via coverage gap.

After computing the values of the conformal loss function associated to each considered score on the test set samples, we calculated the VaR of their distributions by setting $\beta = \alpha$, with $\alpha \in \{0.01, 0.05, 0.1\}$.

Across all datasets, we observed that in every case where marginal coverage was guaranteed, the corresponding $\text{VaR}_\alpha(L_s)$ was constantly zero, independently of the classifier, the score, and the target miscoverage level α . This is globally illustrated in Figure 1, where the horizontal axis shows the coverage gap, i.e., the difference between the achieved empirical coverage and the ideal one ($1 - \alpha$), and the vertical axis reports the $\text{VaR}_\alpha(L_s)$ values. Each point corresponds to one of the experimental configurations across all datasets, models, and error levels, and is shown with blue color if it fulfills the marginal coverage property (i.e., it has a positive coverage gap), or with red color otherwise. Looking at the plot, it is possible to observe that, as the coverage gap is null or positive, then the corresponding VaR_α is zero. This behavior, which empirically confirms Theorem 2, becomes even more intuitive by inspecting the structure of the scores and the associated conformal loss function at the distributional level. To this end, Figure 2 reports a representative example of score and loss distributions for the RF model on `st_landsat` test set, for a fixed $\alpha = 0.1$ ². For each score function, the empirical calibration quantile $q_{1-\alpha}$ induces a separation between prediction sets satisfying or not the coverage guarantees (left and right side from the quantile, respectively). This directly translates into the structure of the conformal loss distribution, which exhibits a peak at $L_s = 0$. As it can be observed in the middle column plots of Figure 2, the empirical VaR_α then collapses to zero across all score functions, while CVaR_α remains sensitive to the magnitude of the tail, yielding, in this example, the lowest value for APS, followed by RAPS, LAC and Margin. Notably, this result is linked to the tail structure of the score distributions as well: the score minimizing CVaR is the one whose distribution presents the narrowest tails (see APS and RAPS scores in Figures 2(g)-2(j)); in contrast, the worst CVaR values come from score distributions with a tail that largely spread away from the quantile (see LAC and Margin in Figures 2(a)-2(d)). The empirical Cumulative Distribution Function (CDF), depicted in the right column plots, provides an even more clear insight on Theorem 2, showing a sharp jump at $L_s = 0$, and a shift towards $L_s > 0$ as soon as the CDF crosses level $1 - \alpha$.

5.2.2. CANONICAL METRICS VERSUS CVAR

The performance of the four score functions has been comparatively evaluated using canonical average coverage and size, alongside the proposed empirical CVaR_β (7) at levels $\beta = \alpha$, $\alpha \in [0.01, 0.1]$.

Figure 3 reports the results obtained for all score functions, datasets and models at $\alpha = 0.05$. Overall, the average coverage satisfies the marginal validity property across all scenarios. We can observe that APS yields frequent overcalibrated results, especially for RF model, with coverage reaching the maximum value of 1 (see the green bars above the black dashed line in the first column of plots), therefore above the expected $1 - \alpha$ threshold. In these situations, the corresponding prediction sets are trivial, as indicated by the average size, which reaches the maximum number of classes of the related datasets. Notably, the corresponding CVaR_α

2. For the sake of conciseness, we just show this analysis for a single dataset, model and α . Completely similar considerations hold for all the other experimental cases.

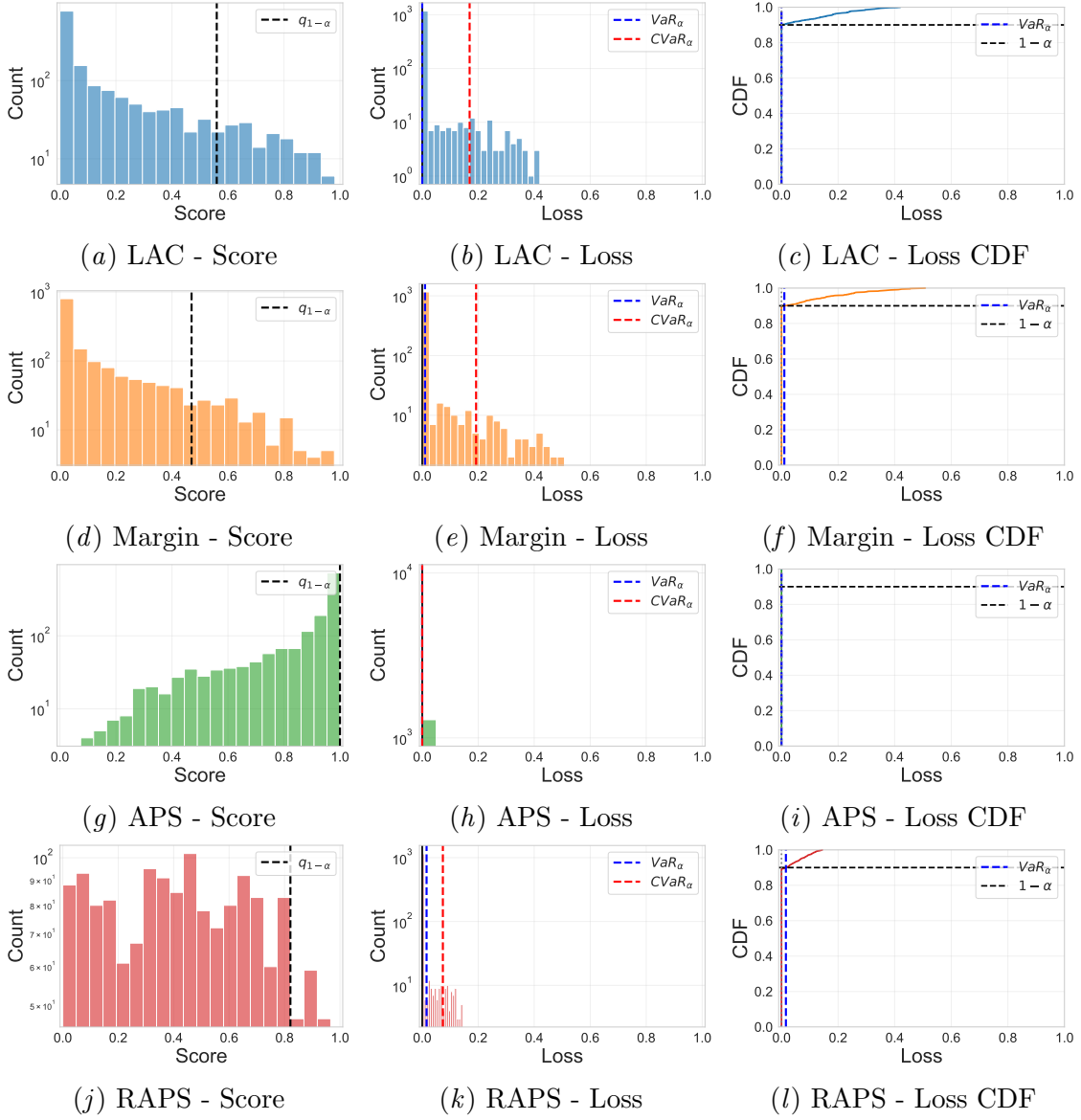


Figure 2: Example of scores and losses distributions on test set (st_landsat dataset, RF model). For each score (first column), the empirical quantile $q_{1-\alpha}$ at $\alpha = 0.1$ is shown with a black dashed line. Loss plots (middle column) report the VaR and CVaR values at $\beta = \alpha$ (blue and red dashed lines, respectively). Notably, $VaR_\alpha = 0$ across all score functions, as also visible from empirical CDFs of the losses (right column).

drops to 0, as no green bar is visible on the rightmost plots. Conversely, more efficient scores such as LAC and margin lead to larger CVaR.

Similar outcomes are observed across other values of α , thus suggesting a trade-off between average set size and $CVaR_\alpha$ that becomes more pronounced as α decreases. Specifically, minimizing the average set size tends to favor overly confident score functions that produce

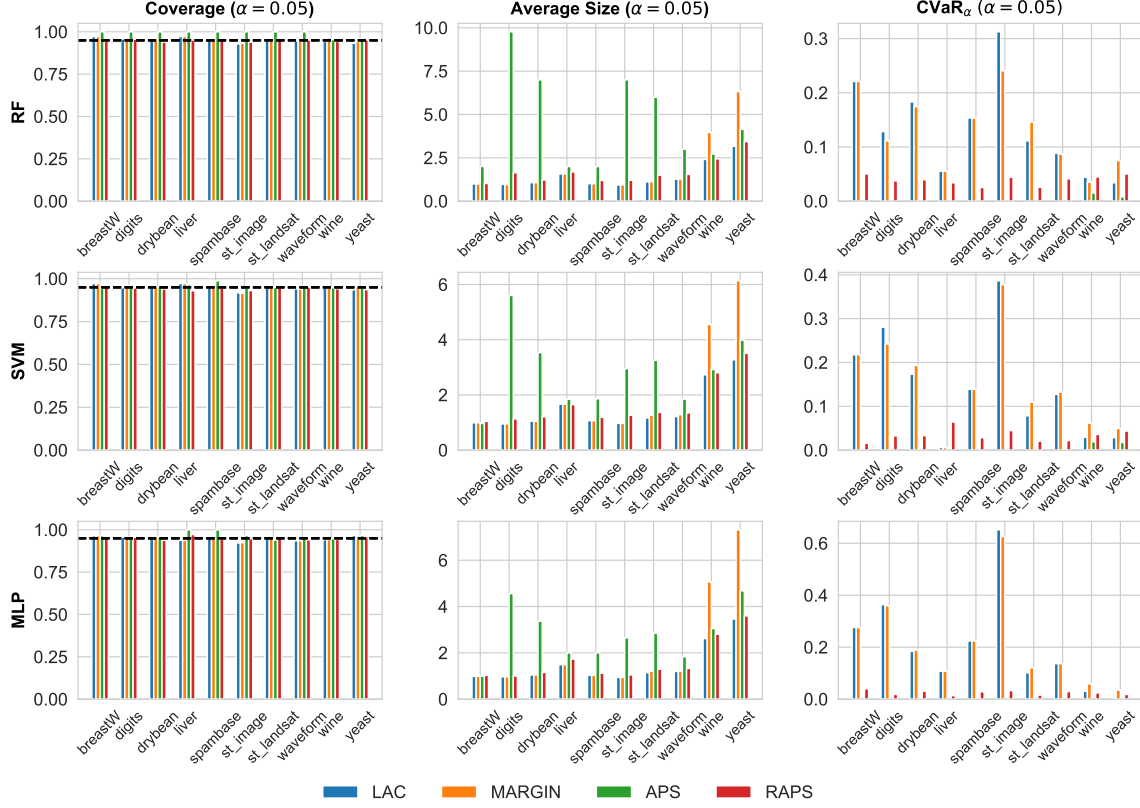


Figure 3: Average coverage, average set size, and CVaR_α values across all datasets and models at $\alpha = 0.05$.

smaller prediction sets though larger CVaR_α , whereas CVaR-based selection favors more conservative score functions that mitigate miscoverage risk.

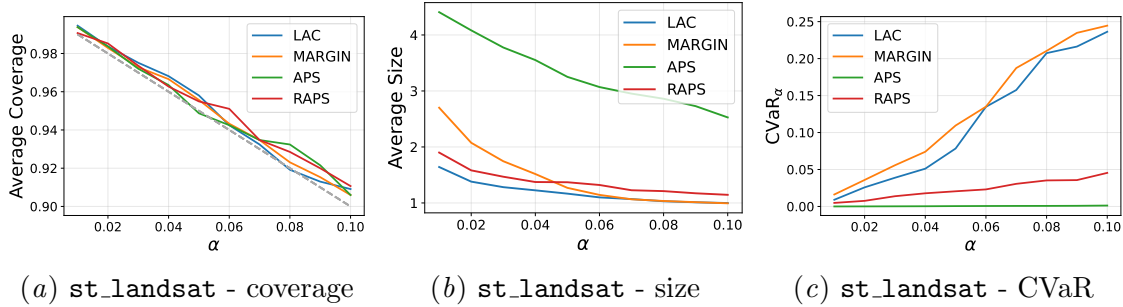


Figure 4: Trend of CP canonical metrics (average coverage and size) at varying $\alpha \in [0.01, 0.1]$, for the `st_andsat` dataset with SVM model.

As an example, we illustrate the full metrics trends with respect to α in Figure 4, using `st_andsat` dataset and SVM model, where it is possible to well observe the opposing behaviour between size and CVaR in selecting the scores, with the conservatism of APS,

Table 2: Comparison of size-based and CVaR-based score selection at $\alpha = 0.05$. For each dataset, the minimum percentage size gap between the two criteria is highlighted in **bold**.

				Size criterion			CVaR criterion			Δ Size [%]
		Best by size	Best by CVaR	Avg Set Size	Coverage	CVaR $_{\alpha}$	Avg Set Size	Coverage	CVaR $_{\alpha}$	
breastW ($n_{\text{classes}} = 2$)	RF	LAC	APS	0.99	0.97	0.22	2.00	1.00	0.00	50.35
	SVM	APS	APS	0.96	0.94	0.00	0.96	0.94	0.00	0.00
	MLP	LAC	APS	0.99	0.96	0.28	0.99	0.96	0.00	0.35
digits ($n_{\text{classes}} = 10$)	RF	MARGIN	APS	0.96	0.95	0.11	9.77	1.00	0.00	88.08
	SVM	LAC	APS	0.95	0.95	0.28	5.60	0.96	0.00	46.54
	MLP	MARGIN	APS	0.97	0.96	0.36	4.56	0.96	0.00	35.93
drybean ($n_{\text{classes}} = 7$)	RF	LAC	APS	1.07	0.95	0.18	7.00	1.00	0.00	84.71
	SVM	MARGIN	APS	1.04	0.95	0.19	3.53	0.95	0.00	35.59
	MLP	LAC	APS	1.04	0.95	0.18	3.37	0.96	0.00	33.20
liver ($n_{\text{classes}} = 2$)	RF	LAC	APS	1.57	0.97	0.06	2.00	1.00	0.00	21.68
	SVM	RAPS	APS	1.65	0.93	0.06	1.85	0.97	0.00	9.79
	MLP	LAC	APS	1.49	0.94	0.11	2.00	1.00	0.00	25.52
spambase ($n_{\text{classes}} = 2$)	RF	LAC	APS	1.01	0.96	0.15	2.00	1.00	0.00	49.56
	SVM	LAC	APS	1.07	0.95	0.14	1.86	0.99	0.00	39.74
	MLP	LAC	APS	1.03	0.95	0.22	2.00	1.00	0.00	48.61
st_image ($n_{\text{classes}} = 7$)	RF	LAC	APS	0.94	0.93	0.31	7.00	1.00	0.00	86.64
	SVM	MARGIN	APS	0.97	0.92	0.38	2.95	0.94	0.00	28.35
	MLP	LAC	APS	0.94	0.92	0.65	2.65	0.97	0.00	24.40
st_landsat ($n_{\text{classes}} = 6$)	RF	LAC	APS	1.11	0.95	0.11	5.99	1.00	0.00	81.29
	SVM	LAC	APS	1.17	0.96	0.08	3.25	0.95	0.00	34.76
	MLP	LAC	APS	1.14	0.95	0.10	2.85	0.94	0.00	28.50
waveform ($n_{\text{classes}} = 3$)	RF	LAC	APS	1.27	0.95	0.09	3.00	1.00	0.00	57.70
	SVM	LAC	APS	1.22	0.94	0.13	1.85	0.96	0.00	21.10
	MLP	LAC	APS	1.20	0.94	0.14	1.83	0.95	0.00	21.03
wine ($n_{\text{classes}} = 7$)	RF	LAC	APS	2.41	0.94	0.04	2.72	0.95	0.02	4.50
	SVM	LAC	APS	2.73	0.95	0.03	2.92	0.95	0.02	2.73
	MLP	LAC	APS	2.62	0.94	0.03	3.05	0.95	0.01	6.18
yeast ($n_{\text{classes}} = 10$)	RF	LAC	APS	3.16	0.93	0.03	4.14	0.96	0.01	9.76
	SVM	LAC	APS	3.27	0.94	0.03	3.99	0.95	0.02	7.14
	MLP	LAC	APS	3.46	0.97	0.01	4.68	0.97	0.00	12.18

generating larger sets, emerging as the best solution from the risk-averse perspective of CVaR. The selection of the best score function then yields different results depending on whether the minimum average set size criterion or the proposed minimum CVaR criterion is adopted. Table 2 reports the best score functions selected according to both criteria, along with the corresponding coverage, average set size, and CVaR metrics, again at $\alpha = 0.05$. It can be observed that in more than 96% of cases the two selection strategies disagree, while they agree for less than 4%. Such disagreement rate increases when α increases (e.g., 100% disagreement with $\alpha = 0.1$), and vice versa (e.g., 90% disagreement with $\alpha = 0.01$). This trend suggests that, as the conformal threshold becomes less conservative, size-based

selection increasingly favors compact prediction sets, while CVaR-based selection remains sensitive to the severity of the induced tail errors.

Notably, under the CVaR-based criterion, APS is consistently selected as the best score function, reflecting its more conservative behavior and its ability to reduce tail-risk severity. The Table also reports the percentage size difference ΔSize , relative to the number of classes, between the two criteria, highlighting (in bold text) the minimum gap obtained. We can observe that, in these cases, such difference spans from 2.73% (**wine** dataset) to 39.74% (**spambase**), with an average of $22.49\% \pm 13.33\%$, thus still suggesting a good compromise between size and CVaR. Conversely, for a few cases (i.e., RF model on **digits**, **drybean** and **st_image**) ΔSize keeps larger than 80%.

5.2.3. CVAR GAPS UNDER SIMILAR COVERAGE AND SET SIZE: A PAIRWISE ANALYSIS

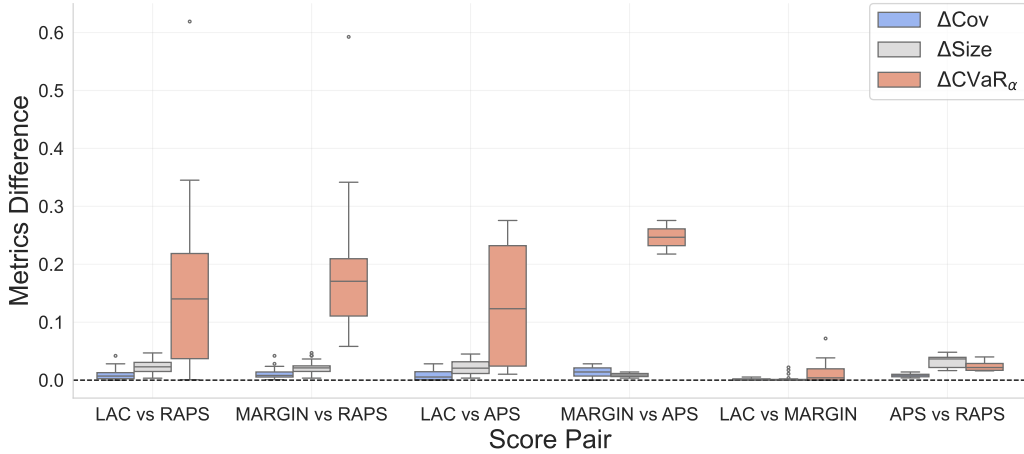


Figure 5: Box plots showing the distributions of coverage, average set size, and CVaR_α absolute variations across all datasets and models, for all possible pair of score functions. Only pairs yielding small variations on standard metrics (i.e., $\Delta\text{Cov} \leq 0.05$ and $\Delta\text{Size} \leq 0.05$) contribute to the plot.

The previous analysis showed that CVaR-based selection often disagrees with the minimum-size criterion, favoring more conservative score functions that reduce tail risk. We now investigate a complementary problem, that is whether CVaR can distinguish between score functions even when the standard metrics, namely coverage and average set size, are nearly identical. In other words, we put the focus of the analysis on those cases where scores only differ in the magnitude of their tail.

To this end, we carried out a pairwise analysis of the outcomes of our experiments across all possible score pairs among the four considered functions, at a fixed level $\alpha = 0.05$. Specifically, for each pair, we computed the absolute differences in average coverage, average set size (normalized with respect to the number of classes), and CVaR_α . We denote these variations as ΔCov , ΔSize , and ΔCVaR_α , respectively. We then selected all score pairs such that $\Delta\text{Cov} \leq 0.05$ and $\Delta\text{Size} \leq 0.05$, i.e., pairs whose coverage and size differ by no more

than 5%. Under this filtering criterion, we retained 76 cases out of the original 180³ (i.e., more than 42%).

Figure 5 shows the distributions of ΔCov , ΔSize , and ΔCVaR_α (with $\alpha = 0.05$) across all datasets and models for each selected score pair after filtering. While variations in coverage and size remain small, as expected from our filtering strategy, ΔCVaR_α exhibits larger variability, especially for pairs involving conservative scores compared with less conservative ones, i.e.: LAC vs APS ($\Delta\text{CVaR}_\alpha = 0.133 \pm 0.133$), LAC vs RAPS ($\Delta\text{CVaR}_\alpha = 0.154 \pm 0.147$), Margin vs APS ($\Delta\text{CVaR}_\alpha = 0.247 \pm 0.041$), and Margin vs RAPS ($\Delta\text{CVaR}_\alpha = 0.200 \pm 0.127$). In contrast, pairs of scores sharing similar tail behavior are characterized by lower CVaR variability (APS vs RAPS: $\Delta\text{CVaR}_\alpha = 0.024 \pm 0.010$; LAC vs Margin: $\Delta\text{CVaR}_\alpha = 0.012 \pm 0.018$). These results indicate that these pairs of score functions are more similar from a risk-averse perspective, although CVaR still provides a preference that is not evident from coverage and average set size alone.

To assess whether these CVaR variations corresponded to a systematic preference for one score over the other, and to identify which one, we further analyzed the direction of the reported CVaR differences. For each pairwise comparison, we retrieved the winning score function as the one with the lowest CVaR_α . For the three pairs involving APS, we found that it consistently outperformed the competing scores in the 100% of times. The remaining pairs were as follows: LAC vs RAPS, with RAPS winning 74% of the times; Margin vs RAPS, with RAPS winning in the 94% of the times, and LAC vs Margin, with Margin winning 75% of the times.

Analyzing the aggregate distribution of winning scores across all available pairwise comparisons, RAPS emerged as the best score in the 43% of the whole cases, followed by LAC (28%), APS (16%), and Margin (13%). This result is driven by the fact that RAPS tends to outperform LAC and Margin in pairwise comparisons, although it is less frequently preferred when compared against APS. It is also worth noting that this aggregate pairwise ranking is not directly comparable with the ranking reported in the end of Section 5.2.2, since it is computed only over filtered pairwise comparisons where coverage and size are similar.

5.3. Key insights

Overall, the findings from our experiments show that CVaR_α provides complementary, risk-averse, information to standard CP evaluation metrics.

Size-CVaR trade-off. The disagreement between size-based and CVaR-based selection (Sec. 5.2.2) highlights a clear trade-off between efficiency and tail-risk mitigation objectives. In our experiments, this disagreement becomes more pronounced as α increases, suggesting that less conservative conformal thresholds make the conflict between compact prediction sets and error control more evident. In particular, size-based selection tends to favor more efficient scores, such as LAC and Margin, whereas CVaR-based selection consistently favors APS, whose more conservative behavior reduces the severity of tail losses. Although average size and CVaR_α generally favor opposite selection criteria, the analysis of percentage size difference (relative to the classes) in Table 2 also suggests that a useful compromise can be identified, with a size gap that keeps lower than the 40% in the 70% of cases.

3. This number results from combining 6 unique score pairs $\times$ 3 models $\times$ 10 datasets.

CVaR enables a finer evaluation level. At the same time, the pairwise analysis (Sec. 5.2.3) shows that CVaR_α remains discriminative even when coverage and average set size are very close across different score functions. This indicates that CVaR is not only an alternative selection criterion, but also an operative tool for identifying hidden differences in the risk profiles of CP algorithms. The filtered pairwise comparisons further show that scores with similar coverage and size may still differ substantially in terms of tail-risk behavior, revealing distinctions that canonical metrics alone would not expose.

Risk-aversion as a new viewpoint. According to these considerations, the proposed CVaR-based evaluation should be regarded as complementary to canonical metrics: coverage measures whether errors occur at the desired frequency, set size measures how informative prediction sets remain under the satisfaction of marginal coverage, and CVaR measures how severe errors are when they occur. In this sense, CVaR should not be interpreted as a replacement for standard CP metrics, but rather as a novel evaluation dimension associated with the severity of worst-case scenarios.

6. Conclusion and future work

In this paper, we proposed a risk-averse perspective for evaluating and selecting conformal score functions through CVaR. Starting from a conformal loss defined as the positive excess of the true-class score over the calibration quantile, we showed that CVaR provides a principled way to quantify the severity of miscoverage events. In particular, when the CVaR tail level is the same of CP error level, CVaR admits a direct interpretation as a measure of miscoverage severity under exact coverage, and as a joint measure of severity and conservativeness under non-exact coverage.

Our experimental analysis confirms that CVaR provides information that is complementary to standard CP metrics. While empirical coverage assesses validity and average set size assesses efficiency, CVaR captures tail error severity. This leads to score function selections that often differ from those obtained by minimizing average set size, especially under more stringent error levels. Moreover, the pairwise analysis shows that CVaR can discriminate among score functions even when coverage and average set size are nearly indistinguishable, revealing differences in the risk profiles of conformal predictors that would otherwise remain hidden.

Future work will investigate the integration of CVaR-based criterion directly into the score function design, rather than using CVaR only as an evaluation and selection tool. Furthermore, other research will be conducted to extend the work to other definitions for the conformal loss functions. Finally, the approach will be also experimented on real-world high-stakes scenarios.

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